Module/Pathway/Taxon Review: L-Tyrosine Catabolism to Fumarate and Acetoacetate in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target pathway:** KEGG ppu00350 "Tyrosine metabolism" (module scope: five-reaction route L-tyrosine → fumarate + acetoacetate) **Module area:** amino-acid metabolism

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1. Executive Summary

The L-tyrosine → fumarate + acetoacetate module is **satisfiable** in *Pseudomonas putida* KT2440. All five expected reactions are encoded in the genome, and the four lower/oxidative steps (steps 2–5) are supported by strong, in some cases direct, target-organism evidence. The route proceeds L-tyrosine → 4-hydroxyphenylpyruvate → homogentisate → 4-maleylacetoacetate → 4-fumarylacetoacetate → fumarate + acetoacetate, and converges on the **homogentisate central catabolic pathway** encoded by the clustered *hmgABC* genes plus *hpd*. These loci map to concrete candidates: **hpd** = **PP_3433**, **hmgA** = **PP_4621** (homogentisate 1,2-dioxygenase), **hmgC** = **PP_4619** (maleylacetoacetate isomerase), and **hmgB** = **PP_4620** (fumarylacetoacetase). This assignment is anchored by detailed genetic and biochemical characterization in the near-identical strain *P. putida* U PMID: 15262943, by whole-genome mapping of the KT2440 aromatic catabolic machinery PMID: 12534466, and by direct proteomic detection of Hpd and HmgA in KT2440 under aromatic-amino-acid catabolic conditions PMID: 16470664.

The **only genuinely ambiguous step is the entry transamination** (step 1), which converts L-tyrosine + 2-oxoglutarate to 4-hydroxyphenylpyruvate + L-glutamate. This activity is present but is carried by a promiscuous, paralog-redundant class-I aminotransferase family rather than by a uniquely demonstrated locus. The best candidates are **tyrB** = **PP_1972** and its equal-length paralog **amaC** = **PP_3590**; both are annotated (by homology/rule propagation) with the exact GO term for L-tyrosine:2-oxoglutarate transaminase activity. Step 1 should therefore be marked **candidate_uncertain / covered-by-redundant-paralogs**, not gap.

A curation-critical observation is that **most of the 16 metadata candidate genes are over-propagated into KEGG ppu00350 and are not part of this module.** Semialdehyde dehydrogenases (davD, sad-I, gabD-II), a formaldehyde dehydrogenase (frmA), a phenylacetaldehyde dehydrogenase (peaE), a short-chain alcohol dehydrogenase (adhP), a histidine-biosynthesis aminotransferase (hisC), a tryptophan monooxygenase (PP_4983), and a DOPA/tyramine-branch decarboxylase (PP_2552) have no mechanistic role in the tyrosine → fumarate + acetoacetate route. Notably, **PP_1709**, annotated as a "fumarylacetoacetate hydrolase family protein," is a **236-aa FAH-superfamily paralog** that is far too short to be a genuine fumarylacetoacetase (the real HmgB/PP_4620 is 430 aa with EC 3.7.1.2) and should not be counted as a second physiological FAH. A further caveat for automated satisfiability scoring: the true module genes carry KEGG buckets **ppu00643** / **ppu00130** / **ppu00401** rather than ppu00350, so the raw ppu00350 metadata *understates* real coverage.

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2. Target-Organism Pathway Definition

2.1 Exact process included

The module comprises the **catabolic conversion of L-tyrosine into two central-metabolism products, fumarate and acetoacetate**, via five enzymatic reactions:

1. **Transamination** — L-tyrosine + 2-oxoglutarate → 4-hydroxyphenylpyruvate + L-glutamate
2. **Oxidative decarboxylation / hydroxylation** — 4-hydroxyphenylpyruvate + O_2 → homogentisate + CO_2
3. **Aromatic ring cleavage** — homogentisate + O_2 → 4-maleylacetoacetate
4. **cis–trans isomerization** — 4-maleylacetoacetate → 4-fumarylacetoacetate
5. **Terminal hydrolysis** — 4-fumarylacetoacetate → fumarate + acetoacetate

Steps 3–5 constitute the **homogentisate central catabolic pathway**, which in *P. putida* is a hub that also receives carbon from L-phenylalanine (via *phh* → Tyr) and from 3-hydroxyphenylacetate. Fumarate re-enters the TCA cycle; acetoacetate is activated to acetoacetyl-CoA and consumed as an acetyl-CoA source.

2.2 Neighboring pathways to keep separate

- **Tyrosine/DOPA decarboxylation branch (Tyr → tyramine/dopamine):** the aromatic-L-amino-acid/DOPA decarboxylase **PP_2552** belongs here, a *different sub-map of ppu00350*, not this module.
- **Phenylalanine hydroxylation (phh):** feeds tyrosine but is an upstream entry route, not part of the five-step module.
- **GABA / succinate-semialdehyde and glutarate-semialdehyde metabolism** (sad-I, gabD-II, davD): distinct semialdehyde dehydrogenase reactions in ppu00350/other maps.
- **Phenylethylamine / phenylacetaldehyde route** (peaE; ppu00360/00643) and **C1/ formaldehyde metabolism** (frmA; ppu00626).
- **Broad KEGG overview maps** (e.g., ppu01100 "metabolic pathways," ppu01110 "biosynthesis of secondary metabolites") should not be used to judge satisfiability.

2.3 Alternate names / database-specific definitions

- KEGG: **ppu00350 "Tyrosine metabolism"** (broad; includes biosynthesis, decarboxylation, and catabolism branches).
- The catabolic core is variously called the **homogentisate pathway**, the **homogentisate ring-cleavage pathway**, or **"hmg/fah/mai" genes** in the KT2440 genome annotation PMID: 12534466.
- MetaCyc: "L-tyrosine degradation I" (homogentisate route).
- Gene-name note: in *P. putida*, the FAH-step gene is **hmgB** (sometimes *fah*) and the isomerase is **hmgC** (sometimes *mai*); this nomenclature swap between databases is a curation hazard.

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3. Expected Step Model and Satisfiability

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 L-tyrosine
      │  (1) transamination  [TyrB / class-I aromatic aminotransferase]
      │      + 2-oxoglutarate → + L-glutamate
      ▼
4-hydroxyphenylpyruvate
      │  (2) HPD, EC 1.13.11.27   [hpd = PP_3433] + O2 → + CO2
      ▼
 homogentisate
      │  (3) HGD ring cleavage, EC 1.13.11.5  [hmgA = PP_4621] + O2
      ▼
4-maleylacetoacetate
      │  (4) MAAI isomerase, EC 5.2.1.2  [hmgC = PP_4619]
      ▼
4-fumarylacetoacetate
      │  (5) FAH hydrolysis, EC 3.7.1.2  [hmgB = PP_4620]
      ▼
 fumarate + acetoacetate  → central metabolism (TCA / acetyl-CoA)
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Step	Reaction	Enzyme / EC	KT2440 locus	Status	Evidence strength
1	Tyr + 2-OG → 4-HPP + Glu	Aromatic aminotransferase, EC 2.6.1.-	tyrB = PP_1972; paralog amaC = PP_3590	**candidate_uncertain** (covered by redundant paralogs)	Family/ rule-based; no locus-unique KT2440 assay
2	4-HPP + O_2 → homogentisate + CO_2	4-hydroxyphenylpyruvate dioxygenase (HPD), EC 1.13.11.27	**hpd** = **PP_3433**	**covered**	Direct KT2440 proteomics + genome mapping
3	Homogentisate + O_2 → 4-maleylacetoacetate	Homogentisate 1,2-dioxygenase (HGD), EC 1.13.11.5	**hmgA** = **PP_4621**	**covered**	Direct KT2440 proteomics; *P. putida* U biochem
4	4-maleylacetoacetate → 4-fumarylacetoacetate	Maleylacetoacetate isomerase (MAAI), EC 5.2.1.2	**hmgC** = **PP_4619**	**covered**	*P. putida* U biochem; genome cluster
5	4-fumarylacetoacetate → fumarate + acetoacetate	Fumarylacetoacetase (FAH), EC 3.7.1.2	**hmgB** = **PP_4620**	**covered**	*P. putida* U biochem; genome cluster

Bottom line: 4 of 5 steps are covered with strong evidence; 1 of 5 (entry transamination) is present but gene-ambiguous. The module is **satisfiable**.

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4. Candidate Genes and Evidence

4.1 High-confidence module genes

hmgA = PP_4621 (Q88E47) — homogentisate 1,2-dioxygenase, EC 1.13.11.5 (step 3). This is the ring-cleaving dioxygenase, the committed step of the central pathway. UniProt carries a curated FUNCTION statement explicitly placing HmgA in phenylalanine/tyrosine degradation. HmgA was **directly detected as an induced protein in KT2440** when cells were grown on phenylalanine PMID: 16470664, and the enzyme is biochemically characterized in *P. putida* U as part of the *hmgABC* transcriptional unit PMID: 15262943. **High confidence; no caveats.**

hmgB = PP_4620 (Q88E48) — fumarylacetoacetase, EC 3.7.1.2, 430 aa (step 5). The terminal hydrolase releasing fumarate and acetoacetate. It sits immediately adjacent to *hmgA* and *hmgC* in a single operon in *P. putida* U PMID: 15262943. The 430-aa length and assigned EC 3.7.1.2 are consistent with a genuine FAH. **High confidence.**

hmgC = PP_4619 (Q88E49) — maleylacetoacetate isomerase, EC 5.2.1.2 (step 4). A GST-zeta-family isomerase converting the cis (maleyl) to trans (fumaryl) isomer; glutathione-dependent. Part of the *hmgABC* operon PMID: 15262943. **High confidence.**

hpd = PP_3433 (Q88HC7) — 4-hydroxyphenylpyruvate dioxygenase, EC 1.13.11.27 (step 2). Produces homogentisate from 4-hydroxyphenylpyruvate. Directly detected as **induced in KT2440 on phenylalanine** PMID: 16470664 and mapped to the KT2440 chromosome as *hpd* PMID: 12534466. Note KEGG assigns *hpd* to bucket ppu00130 (ubiquinone/other), not ppu00350 — a database-boundary artifact, not biology. **High confidence.**

4.2 Entry-step candidates (transamination, step 1)

tyrB = PP_1972 (Q88LG1) — aromatic-amino-acid aminotransferase, 398 aa, EC 2.6.1.- . Assigned the entry reaction (Tyr → 4-hydroxyphenylpyruvate) by analogy to *P. putida* U TyrB PMID: 15262943. UniProt places it in the class-I PLP-dependent aminotransferase family, with Asp_trans (IPR000796) and class-I PLP site (IPR004838) domains, and annotates GO:0004838 "L-tyrosine:2-oxoglutarate transaminase activity." **Caveat:** its primary KEGG bucket is ppu00401 (aromatic amino-acid *biosynthesis*), and the annotation is rule-/homology-based, not a KT2440 enzyme assay.

amaC = PP_3590 (Q88GX7) — aminotransferase, 398 aa, EC 2.6.1.- . An **equal-length paralog of tyrB** carrying **word-for-word identical** UniProt annotations: same class-I PLP family membership, identical InterPro set (IPR000796, IPR004838), and the identical GO:0004838 +

GO:0030170 (PLP binding) + "L-phenylalanine biosynthesis from chorismate via phenylpyruvate" process term. This identity is the signature of **UniRule/rule-based propagation**, not two independent experimental characterizations. The two loci therefore represent **paralog redundancy**: the transamination activity is present, but neither locus is uniquely demonstrated as *the* physiological Tyr aminotransferase in KT2440.

hisC = **PP_0967 (Q88P86)** (histidinol-phosphate aminotransferase, EC 2.6.1.9) shares EC 2.6.1.- space but belongs to histidine biosynthesis; it is not a serious step-1 candidate.

4.3 Over-propagated / non-module candidates

Locus	Annotation	Real pathway	Verdict
PP_1709 (Q88M65)	FAH-family protein, **236 aa**, no EC, no gene name	FAH superfamily, different substrate	**Not a second FAH**; too short vs 430-aa HmgB
PP_2552 (Q88JU5)	DOPA/aromatic-L-amino-acid decarboxylase, EC 4.1.1.28	Tyr → tyramine decarboxylation branch	Different sub-map; **exclude**
davD = PP_0213	Glutarate-semialdehyde DH, EC 1.2.1.-	Lysine/glutarate	Exclude
sad-I = PP_2488	Succinate-semialdehyde DH, EC 1.2.1.24	GABA/succinate	Exclude
gabD-II = PP_4422	SSADH (NADP+), EC 1.2.1.79	GABA	Exclude
frmA = PP_1616	S-(hydroxymethyl)glutathione DH, EC 1.1.1.1/284	C1/formaldehyde	Exclude
peaE = PP_3463	Phenylacetaldehyde DH, EC 1.2.1.39	Phenylethylamine (ppu00360/643)	Exclude
adhP = PP_3839	Short-chain alcohol DH, EC 1.1.1.-	Generic redox	Exclude
PP_4983 (Q88D45)	Tryptophan 2-monooxygenase, EC 1.13.12.3	Tryptophan	Exclude

PP_1709 in detail: UniProt lists it as a "Fumarylacetoacetate hydrolase family protein," 236 aa, **no gene name and no EC number**. Genuine fumarylacetoacetase (HmgB/PP_4620) is 430 aa with EC 3.7.1.2. The ~200-aa truncation and absent EC strongly indicate PP_1709 is an FAH-*superfamily* member acting on a different substrate (e.g., an acylpyruvate- or oxaloacetate-type hydrolase), **not** a second physiological fumarylacetoacetase. It should not count toward step 5.

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5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 The one real ambiguity — entry transamination

The transamination step is the sole point of genuine uncertainty. It is **not a gap** (activity is annotated on ≥2 loci with the correct GO term) but it is **candidate_uncertain**: aromatic aminotransferases are notoriously promiscuous and functionally redundant, and no KT2440-specific enzyme assay isolates a single physiological Tyr aminotransferase. In *P. putida* U the reaction is attributed to TyrB PMID: 15262943; transfer to KT2440 is reasonable at the *family* level but weak at the *locus* level.

5.2 KEGG bucket mismatch (satisfiability-scoring hazard)

The true module genes are **not** all filed under ppu00350: - *hpd* → ppu00130 - *hmgA/hmgB/hmgC* → ppu00643 (in the local metadata) - *tyrB/amaC* → ppu00401

An automated check that counts only ppu00350-bucketed genes would **under-count** coverage and could falsely flag the module as a gap. The module is satisfied; the bucketing is a database-boundary artifact.

5.3 Over-annotation into ppu00350

Of the 16 candidates, only ~5–6 are mechanistically relevant (hpd, hmgA, hmgB, hmgC, and one of tyrB/amaC). The remaining ~10 loci — semialdehyde dehydrogenases, formaldehyde/alcohol dehydrogenases, a His aminotransferase, a Trp monooxygenase, and the DOPA decarboxylase — are **over-propagated** into the tyrosine-metabolism bucket by broad EC/GO mappings and should be excluded from module satisfiability.

5.4 Broad EC/GO mappings

- EC 2.6.1.- (unspecified aminotransferase) on tyrB, amaC, hisC is too broad to disambiguate function.
- The FAH-superfamily annotation on PP_1709 is a classic broad-family over-call.
- GST-zeta on hmgC and dioxygenase superfamilies on hpd/hmgA are specific enough here to be reliable.

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6. Module and GO-Curation Recommendations

Step	Recommended module status	Locus	Rationale
1. Transamination	**candidate_uncertain**	tyrB = PP_1972 (primary), amaC = PP_3590 (redundant)	Activity present via rule-based GO:0004838 on two paralogs; no locus-unique assay
2. HPD	**covered**	hpd = PP_3433	Direct KT2440 proteomics + genome mapping
3. HGD	**covered**	hmgA = PP_4621	Direct KT2440 proteomics; *P. putida* U biochem
4. MAAI	**covered**	hmgC = PP_4619	*P. putida* U biochem; operon cluster
5. FAH	**covered**	hmgB = PP_4620	*P. putida* U biochem; operon cluster

Additional recommendations:

1. **Do not count PP_1709 toward step 5.** Reannotate as an FAH-superfamily hydrolase of unspecified substrate; flag as "likely over-propagated fumarylacetoacetase annotation."
2. **Remove the ~10 non-module candidates** (davD, sad-I, gabD-II, frmA, peaE, adhP, hisC, PP_4983, PP_2552) from module satisfiability scoring for the tyrosine→fumarate route. PP_2552 should instead anchor a *separate* Tyr-decarboxylation module.

3. **Module boundary is correct for this organism**; no `module_needs_revision`. But the **KEGG-bucket-based candidate harvesting is misleading** — recommend that satisfiability logic ignore the ppu00350 bucket label and instead match on EC / GO / ortholog identity, which correctly picks up hpd (ppu00130) and hmgABC (ppu00643).
4. **GO curation:** hmgA/hmgB/hmgC/hpd already carry appropriate GO MF terms. For step 1, retain GO:0004838 on tyrB and amaC but tag the evidence code as inferred from electronic/rule-based annotation (IEA/UniRule), not experimental.
5. **No new GO term requests appear necessary**; the reactions all have existing EC/GO coverage.

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7. Genes to Promote to Full `fetch-gene` Review

Gene	Locus	Why promote
tyrB	PP_1972	Resolve the entry-step ambiguity; determine whether it is the physiological Tyr aminotransferase vs amaC. Highest-value open question.
amaC	PP_3590	Equal-length paralog with identical rule-based annotation; needs disambiguation from tyrB.
PP_1709	PP_1709	Confirm it is NOT a fumarylacetoacetase; reclassify the 236-aa FAH-superfamily protein and prevent over-count of step 5.
hmgB	PP_4620	Confirm EC 3.7.1.2 assignment and operon membership in KT2440 (currently transferred from *P. putida* U).

hpd, hmgA, and hmgC are well-supported and are **lower priority** for full review, though confirming the KT2440 *hmgABC* operon structure directly (rather than by transfer from strain U) would close a minor gap.

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8. Evidence Base

PMID	Title (abbrev.)	Organism	How it supports the review
15262943	*The homogentisate pathway: a central catabolic pathway...in Pseudomonas putida*	*P. putida* U	Defines steps 3–5 (HmgA/HmgB/HmgC → fumarate + acetoacetate), the *hmgABC* operon, HmgR regulation, and the TyrB entry step. Strongest mechanistic anchor; transfer to KT2440 strong (near-identical strain).
16470664	*Analysis of aromatic catabolic pathways in P. putida KT2440...proteomic approach*	**KT2440 (direct)**	Direct proteomic detection of **Hpd and HmgA induced on phenylalanine** — target-organism evidence for steps 2 and 3.
12534466	*Genomic analysis of the aromatic catabolic pathways from P. putida KT2440*	**KT2440 (direct)**	Whole-genome mapping placing the homogentisate pathway (**hmg/fah/mai**) plus **phh, hpd** on the KT2440 chromosome — confirms genomic presence of the module.

Verbatim supporting quotes:

- PMID: 15262943: *"Homogentisate is then catabolized by a central catabolic pathway that involves three enzymes, homogentisate dioxygenase (HmgA), fumarylacetoacetate hydrolase (HmgB), and maleylacetoacetate isomerase (HmgC), finally yielding fumarate and acetoacetate."* — establishes enzymes and products of steps 3–5.
- PMID: 16470664: *"Phenylalanine induced 4-hydroxyphenyl-pyruvate dioxygenase (Hpd) and homogentisate 1,2-dioxygenase (HmgA), key enzymes in the homogentisate degradation pathway."* — direct KT2440 expression evidence for Hpd (step 2) and HmgA (step 3).
- PMID: 12534466: *"the homogentisate pathway (hmg/fah/mai genes)"* and *"phenylalanine and tyrosine (phh, hpd)"* — confirms genomic presence and correct pathway attribution.

9. Limitations and Knowledge Gaps

1. **Locus-level transamination assignment is unresolved.** Family evidence is strong; the specific physiological gene (tyrB vs amaC vs both) is not experimentally pinned in KT2440.
2. **Steps 4–5 rest partly on transfer from *P. putida* U**, not direct KT2440 enzyme assays. Strain U and KT2440 are closely related, making transfer strong, but the KT2440 *hmgABC* operon structure and MAAI/FAH activities have not been individually re-characterized in KT2440 to the same depth.
3. **PP_1709's true substrate is unknown.** It is confidently *not* the physiological FAH, but its actual role is uncharacterized.
4. **No transcriptomic/mutant fitness data were analyzed here**; the review rests on genome, proteome, and biochemical literature plus UniProt structural/annotation evidence.
5. **Annotation-propagation risk:** the identical UniProt annotations on tyrB/amaC are a strength for "activity present" but a weakness for locus specificity — they reflect a rule, not two experiments.

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10. Proposed Follow-up Experiments / Actions

1. **Single and double knockouts of PP_1972 (tyrB) and PP_3590 (amaC)** in KT2440, followed by growth on L-tyrosine as sole carbon/nitrogen source, to test which paralog(s) support entry transamination and whether they are redundant.
2. **In vitro assay** of purified PP_1972 and PP_3590 for L-tyrosine:2-oxoglutarate transaminase activity (K_m/k_cat), to quantify physiological relevance.
3. **Biochemical characterization of PP_1709** across candidate FAH-superfamily substrates (acylpyruvates, oxaloacetate) to reclassify it and remove it from the FAH step.
4. **Direct verification of the KT2440 hmgABC operon** (RT-PCR/RNA-seq) and its induction by homogentisate, confirming HmgR-type regulation transferred from *P. putida* U.
5. **Curation actions:** update the module to mark steps 2–5 covered and step 1 candidate_uncertain; strip the ~10 over-propagated ppu00350 candidates from satisfiability scoring; add a note that satisfiability logic should match on EC/GO/ortholog identity rather than the ppu00350 KEGG bucket.

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*Prepared for manual module satisfiability and gene-annotation curation. Evidence is explicitly labeled as direct-KT2440, transferred-from-*P. putida*-U, or rule-based homology throughout.*
